

计算机应用数学

CSMATH 2026 · 课程论文要求

浙江大学计算机科学与技术学院 | 2026 春夏学期

面向博士生一年级 · English version attached

一句话要点：本论文环节本质上是一次科研小训练，也是对课程学习成效的综合检验。它考察三件事：能否把一个实际问题形式化、能否对数学对象建立直观、能否把数学方案落成可计算的算法。

截止日期：2026 年 6 月 1 日（周一）24:00。

1 课程立意与论文理念

本课程面向计算机及相关专业一年级博士生，以多元统计、非线性优化、偏微分方程、应用泛函分析四大主题为骨干，配合图形学、视觉、机器学习与数据科学中的真实案例，训练研究生面对未知问题时的**数学思考习惯**。

贯穿全课的方法论，可以概括为三条能力主线：

(1) 问题的形式化 (Formalization)：

把一个含糊的应用问题，剥出其数学骨架——变量、约束、目标、假设、度量空间，用严格的符号写出来。”说清楚你到底在解什么”。

(2) 数学的直观化 (Intuition)：

对抽象对象——分布、流形、对偶、梯度、算子——建立几何或物理图像，能用一句话、一张图说明”它究竟在做什么”，而不止于推导正确。

(3) 数学化与可计算 (Computability)：

把数学方案转成可运行、可复现、可度量的算法；关注复杂度、数值稳定性、收敛条件、工程取舍。”能算出来的数学才是有用的数学。”

课程论文不是”写一篇综述交差”，而是在一个自选的小课题上，**完整走一遍「形式化 → 直观化 → 数学化 → 可计算」**的闭环。这也是未来博士研究反复要做的事。

2 选题范围

2.1 选题原则

- 与课程数学方法有明确关联：所用的核心工具应能在课程讲义中找到对应章节，或是其自然延伸；

- **小而深胜过大而空**：一个你能推导清楚并跑出结果的小问题，远比一篇泛泛而谈的大综述有价值；
- 允许由实际研究问题出发，但必须提炼出其中的数学内核并加以分析。

2.2 推荐方向（2026 年度）

鼓励紧扣课程内容与当前研究前沿，推荐方向包括但不限于：

1. **课程核心方法的深化**：PCA / ICA / 流形学习、EM 与变分推断、图模型推断、线性/二次规划、对偶方法、梯度法 / 共轭梯度 / 拟牛顿、Level-set 与变分方法——任选其一，结合一个应用场景做出可运行的示例；
2. **大语言模型中的数学**：Transformer 注意力的数学刻画、扩展律（scaling law）、位置编码的几何意义、RLHF / DPO 的优化视角；
3. **深度学习在 CG / CV / MM 中的方法**：扩散模型、NeRF、3D Gaussian Splatting、多模态对齐背后的概率与优化；
4. **LLM-based Agent 与推理**：思维链、工具调用、规划与决策中的图模型 / 马尔可夫 / 强化学习视角；
5. **跨学科交叉**：AI4Science、科学计算、具身智能、AIGC 中的数学问题。

2.3 不鼓励的选题

- 与课程数学主题无关的纯产品 / 工程报告；
- 对单一论文的逐段翻译，没有形式化、没有直观解释、没有动手实验；
- 由 AI 生成后未加审阅、看不出作者思考痕迹的稿件。

3 论文结构与内容要求

论文建议按下列结构组织，每一部分都与课程强调的三条能力相呼应：

章节	内容要点（括号内为对应能力）
引言	研究问题、应用背景、本文尝试回答什么。(形式化的起点)
问题形式化	明确给出变量、假设、目标函数、约束、所在空间；指出问题的类别（凸 / 非凸、连续 / 离散、确定 / 随机等）。(形式化)
数学基础与直观	用自己的话解释所涉及的核心概念（一张图、一个小例子胜过十行公式）；点明它与课程哪一章节相关。(直观化)
方法剖析	推导、假设、收敛性或误差分析；必要时讨论复杂度与数值稳定性。(数学化 → 可计算)
实现与实验	自行实现核心算法，设计至少一组可复现实验，展示方法的行为、边界与失败案例。(可计算)
个人见解	方法的优缺点、适用边界、与其他方法的比较、可能的改进方向。(研究训练的核心)
参考文献	BibTeX 管理，>5 条高质量引用，优先顶会 / 顶刊与 arXiv。

3.1 基础部分（80%）：理解、形式化、直观化

目标：选定主题或代表作，用自己的语言讲清楚“问题是什么、数学长什么样、为什么这样做行”。拒绝仅做翻译的搬运。

3.2 进阶部分（20%）：数学化与可计算

目标：把论文里的数学“跑起来”。

- 自行实现核心算法（可小规模），给出**可执行文件、运行说明**与小样本示例；
- 至少一张图 / 一张表汇报实验结果；
- 鼓励复现、小规模创新或对比实验；
- 推荐使用 **Python**；C++ / Julia / Matlab 亦可，但须保证可复现。

AI 辅助使用说明：允许使用 ChatGPT、Claude 等工具辅助写作与编码，但须在致谢节如实说明使用范围（润色语言 / 辅助推导 / 生成代码骨架等）。**核心的形式化、推导、直观解释、实验结论必须由本人独立完成并理解。**

4 格式规范

项目	要求
篇幅	正文 6–12 页 (A4)，不含封面、参考文献与附录
模板	推荐 LaTeX (IEEE / ACM / Elsevier 等通用模板皆可)
文字	中英文皆可，语言一致、术语规范
字体	正文 10–11pt；标题层级清晰
数学公式	使用 LaTeX 排版，不接受截图公式
图表	矢量图优先；每张图表需有编号、标题与文中引用
参考文献	BibTeX 管理，>5 条，优先顶会顶刊与 arXiv
文件格式	仅接受 PDF，MS Word 不推荐
命名规则	CSMATH2026_ 学号 _ 姓名 .pdf (代码打包为同名 zip)

5 评分标准

评分维度与课程强调的三条能力一一对应，确保论文确为学习成效的检验：

维度	权重	考察点
问题形式化	20%	变量 / 假设 / 目标 / 约束是否清晰严格
数学直观	20%	能否用图 / 例 / 比喻讲明白抽象对象
方法与推导	20%	推导严谨性、收敛 / 误差 / 复杂度分析
实现与实验	20%	代码可复现、实验设计与结果解读
见解与表达	15%	批判思考、与课程内容的呼应、写作规范
文献广度	5%	引用质量与覆盖面
合计	100%	

6 资源与文献推荐

6.1 会议与期刊

- 图形学 / 视觉：ACM SIGGRAPH, Eurographics, SCA, Pacific Graphics; ICCV, ECCV, ACCV, CVPR;
- 机器学习 / AI：NeurIPS, ICML, ICLR, IJCAI, UAI, AAAI;

- 数据 / 数据库: SIGMOD, SIGKDD, VLDB;
- 预印本: arXiv (cs.LG, cs.CV, cs.CL, math.OC 等)。

6.2 工具与平台

- 文献管理: [Zotero](#) (推荐) / [Mendeley](#);
- 代码托管: GitHub / GitLab, 建议公开仓库并附 README;
- 课程资料: github.com/hongxin/csmath。

7 关键日期

节点	日期
论文与代码提交截止	2026-06-01 (周一) 24:00
评阅与成绩反馈 (预计)	2026 年暑期, 不晚于 2026-07-31

「再，斯可矣。」——《论语》
思考够了，就去动手。

Applied Mathematics for Computer Science

CSMATH 2026 · Course Paper Requirements

School of Computer Science & Technology, Zhejiang University | Spring–Summer 2026

For first-year PhD students · English version for international students

In one line: This course paper is a **small research exercise** and a comprehensive check on what you have learned. It evaluates three things: **whether you can formalize a real problem, whether you can develop intuition for mathematical objects, and whether you can turn a mathematical solution into a computable algorithm.**

Deadline: 2026-06-01 (Mon) 24:00.

1 Course Philosophy and the Role of the Paper

This course targets **first-year PhD students in computer science and related areas**. Built around four pillars—**multivariate statistics, nonlinear optimization, partial differential equations, and applied functional analysis**—it uses real cases from graphics, vision, machine learning and data science to cultivate a **habit of mathematical thinking** when facing unfamiliar problems.

The methodology running through the whole course can be summarized as three core abilities:

(1) Formalization of problems.

Strip a vague application down to its mathematical skeleton—variables, constraints, objectives, assumptions, underlying space—and write it precisely. *"Say exactly what you are solving."*

(2) Intuition for mathematics.

Build geometric or physical images of abstract objects—distributions, manifolds, duality, gradients, operators—so that you can explain *what they really do* in one sentence or one picture, not merely derive correctly.

(3) Mathematization and computability.

Translate the mathematical formulation into an algorithm that actually runs, is reproducible and measurable; pay attention to complexity, numerical stability, convergence and engineering trade-offs. *"Math you can compute is useful math."*

The course paper is *not* "a survey handed in to finish a task." It is a complete round-trip of **Formalize** → **Intuit** → **Mathematize** → **Compute** on a self-chosen small topic—the very loop you will repeat throughout your PhD research.

2 Topic Selection

2.1 Principles

- **Clearly related to the mathematical methods of the course:** the core tools you use should map to a specific course chapter or be its natural extension;
- **Small and deep beats broad and shallow:** a small problem you can derive clearly and run to results is worth more than a vague mega-survey;
- You may start from a real research problem, but you must extract and analyze its mathematical core.

2.2 Recommended Directions (2026)

1. **Deepening core course methods:** PCA / ICA / manifold learning, EM and variational inference, graphical-model inference, LP / QP, duality, gradient / conjugate-gradient / quasi-Newton methods, level-set and variational methods—pick one and build a runnable example around an application.
2. **Mathematics behind large language models:** attention mechanisms, scaling laws, geometry of positional encodings, RLHF / DPO from an optimization viewpoint.
3. **Deep learning in CG / CV / MM:** diffusion models, NeRF, 3D Gaussian Splatting, multi-modal alignment—their probabilistic and optimization underpinnings.
4. **LLM-based agents and reasoning:** chain-of-thought, tool use, planning and decision-making via graphical models / Markov processes / reinforcement learning.
5. **Cross-disciplinary:** AI4Science, scientific computing, embodied intelligence, AIGC-related mathematical questions.

2.3 Discouraged Topics

- Pure product / engineering reports unrelated to the course mathematics;
- Line-by-line translation of a single paper with no formalization, no intuition, no experiments;
- AI-generated drafts with no visible trace of the author's own thinking.

3 Structure and Content

Each section of the paper should echo one of the three course abilities:

Section	Content (ability in parentheses)
Introduction	research question, background, what the paper tries to answer. (<i>start of formalization</i>)
Problem formalization	state variables, assumptions, objective, constraints, space; classify the problem (convex / non-convex, continuous / discrete, deterministic / stochastic). (<i>formalization</i>)
Math background & intuition	explain core concepts in your own words; a picture or tiny example beats ten lines of formulae; point to the matching course chapter. (<i>intuition</i>)
Method analysis	derivation, assumptions, convergence / error analysis; complexity and stability where relevant. (<i>mathematization</i> $\rightarrow$ <i>computability</i>)
Implementation & experiments	implement the core algorithm yourself; at least one reproducible experiment showing behavior, limits and failure cases. (<i>computability</i>)
Personal insights	pros, cons, applicability, comparisons, possible improvements. (<i>the research-training core</i>)
References	BibTeX, >5 high-quality entries; prefer top venues and arXiv.

3.1 Basic Part (80%): Understanding, Formalization, Intuition

Pick a topic or landmark paper and explain, in your own words, *what the problem is, what the math looks like, and why the method works*. Do not just transcribe.

3.2 Advanced Part (20%): Mathematization & Computability

Make the math *run*:

- implement the core algorithm (small scale is fine), ship the **executables and run instructions**, with a small sample;
- at least one figure / table to report results;
- reproduction, small-scale innovation, or comparative experiments are welcomed;
- **Python** is recommended; C++ / Julia / Matlab are fine if reproducible.

Note on AI assistance. ChatGPT, Claude and similar tools may be used to help with writing and coding, but you must disclose the scope in the acknowledgments (e.g., language polishing, derivation assistance, code scaffolding). **The formalization, derivations, intuitions and experimental conclusions must be understood and produced by you.**

4 Format

Item	Requirement
Length	6–12 pages (A4), excluding cover, references, appendix
Template	LaTeX preferred (IEEE / ACM / Elsevier styles all accepted)
Language	Chinese or English, internally consistent
Font	10–11 pt body; clear heading hierarchy
Math	typeset with LaTeX; screenshot formulas not accepted
Figures	vector preferred; numbered, captioned, cited in text
References	BibTeX, >5 entries, top venues / arXiv preferred
File format	PDF only ; M\$ Word <i>not</i> recommended
Naming	CSMATH2026_StudentID_Name.pdf (code as same-name zip)

5 Grading

Dimensions mirror the three course abilities, so the paper truly reflects what you have learned:

Dimension	Weight	Focus
Formalization	20%	clarity / rigor of variables, assumptions, objective, constraints
Mathematical intuition	20%	explaining abstract objects via figures, examples, analogies
Method & derivation	20%	rigor, convergence / error / complexity analysis
Implementation & exp.	20%	reproducibility, experiment design, result interpretation
Insight & writing	15%	critical thinking, tie to course material, writing quality
References	5%	quality and breadth of citations
Total	100%	

6 Resources

- **Graphics / Vision:** ACM SIGGRAPH, Eurographics, SCA, Pacific Graphics; ICCV, ECCV, ACCV, CVPR;
- **ML / AI:** NeurIPS, ICML, ICLR, IJCAI, UAI, AAAI;
- **Data / DB:** SIGMOD, SIGKDD, VLDB;
- **Preprints:** arXiv (cs.LG, cs.CV, cs.CL, math.OC, ...);
- Reference management: [Zotero](#) (recommended) / [Mendeley](#);
- Code hosting: GitHub / GitLab with a README;
- Course materials: github.com/hongxin/csmath.

7 Key Dates

Milestone	Date
Paper & code submission deadline	2026-06-01 (Mon) 24:00
Grading & feedback (expected)	summer 2026, no later than 2026-07-31

“Twice is enough.” — Analects
Once you have thought enough, get to work.